

 Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)		COURSE SYLLABUS	
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(DS)		W.E.F	AY: 2025 - 2026 (Rev. 2023 (NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY CSE(DS)		COURSE NAME	Applied Mathematics
		COURSE CODE	2301259T
		COURSE CREDITS	3
RELEASE DATE : 01/07/2025		REVISION NO.	0.0

TEACHING SCHEME		EXAMINATION SCHEME AND MARKS					
(HOURS/WEEK)		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
3	NIL	30	20	50	NIL	NIL	100

PREREQUISITE KNOWLEDGE: Basics of Mathematics

COURSE OBJECTIVES :

- 2301254T.CEO.1: To provide mathematical and statistical tools essential for data science and machine learning.
- 2301254T.CEO.2: To build a solid foundation in linear algebra, probability, optimization, and data-driven inference.
- 2301254T.CEO.3: To apply practical statistical techniques to real-world data problems using Python.

COURSE OUTCOMES :

After successful completion of the course, students will be able to,

- 2301254T.CO.1: Apply concepts of linear algebra such as vectors, matrices, eigenvalues, and SVD in the context of data transformations and dimensionality reduction (e.g., PCA) [L3]
- 2301254T.CO.2: Make use of probability theory and statistical distributions to model uncertainty in data-driven systems [L3].
- 2301254T.CO.3: Select appropriate data summarization, sampling, and hypothesis testing to make inferences and support decisions based on data [L4].
- 2301254T.CO.4: Construct and evaluate regression models and apply information-theoretic concepts like entropy and MLE in machine learning contexts [L4].
- 2301254T.CO.5: Implement gradient-based optimization techniques to observe the convergence behavior [L4].
- 2301254T.CO.6: Make a choice of appropriate statistical, mathematical, and visualization techniques using Python tools to explore and interpret real-world datasets [L4].

COURSE ABSTRACT:

This course introduces core concepts of data preprocessing, statistical analysis, regression, and time series forecasting for data-driven decision-making. It covers data wrangling techniques, descriptive and inferential statistics, hypothesis testing, regression models, correlation analysis, and time series methods like ARIMA and Prophet. Real-world case studies from retail, healthcare, real estate, and stock markets are used to build analytical and predictive modeling skills.

THEORY COURSE CONTENTS

UNIT 1	Linear Algebra and Matrix Computations	06 HOURS
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Applications/Case Study: Principal Component Analysis (PCA), Neural Network weight transformations

Contents: Vectors, vector spaces, Scalars, vectors, matrices, tensors, Matrix operations: addition, multiplication, inverse, Rank, null space, orthogonality, Eigenvalues and eigenvectors, Singular Value Decomposition (SVD).

Self Study: Gaussian Elimination

Further Readings: Matrix Decomposition

UNIT 2	Probability and Random Variables	08 HOURS
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Applications/Case Study: Covid Data Analysis or similar

Contents: Probability definitions: Classical, empirical, axiomatic, Conditional probability, Bayes' Theorem, Discrete and continuous random variables, Common distributions: Bernoulli, Binomial, Poisson, Normal, Exponential, Expectation, variance, standard deviation, Covariance and correlation

Self Study: variance, standard deviation Common distributions: Bernoulli, Binomial, Poisson, Normal, Exponential

Further Readings: multivariate Central Limit theorem

UNIT 3	Statistics, Sampling, and Visualization	08 HOURS
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Applications/Case Study: Covid Data Analysis or similar

Contents: Descriptive statistics: Measures of central tendency and dispersion, Skewness and kurtosis, Outlier detection, data transformation, Normalization and standardization, Sampling techniques: Simple random, stratified, cluster, systematic, Central Limit Theorem, Sample size calculation, margin of error, Sampling techniques and bias, Confidence intervals, point estimation, Hypothesis testing: Z-test, T-test, Chi-square, ANOVA, F test, Type I II errors, p-values, A/B testing and interpretation.

Self Study: Data Visualization, F test, T-test

Further Readings: A/B testing and interpretation

UNIT 4	Regression and Information Theory	08 HOURS
Applications/Case Study: Contents: Simple and multiple linear regression, Bias-variance tradeoff, Underfitting vs. Overfitting, Polynomial regression, Assumptions and diagnostics, Logistic regression (overview), Information Gain, Entropy, cross-entropy, KL divergence, Naïve Bayes classifier, Maximum Likelihood Estimation, Gaussian Mixture Models (GMMs), MAP estimation, Naïve Bayes and Gaussian classifier, Regularization techniques: L1, L2, Cross-validation methods, Evaluation metrics for ML/DL models: Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC, Regression: RMSE, MAE, R^2 Self Study: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Regression: RMSE, MAE, R^2, Further Readings: Evaluation metrics for ML/DL Model		
UNIT 5	Calculus and Its Role in Deep Learning	08 HOURS
Applications/Case Study: Contents: functions, limits, derivatives, . Dot product (projection, similarity), Cross product (3D Applications), Partial derivatives and gradients, Chain Rule (crucial for backpropagation), Jacobian and Hessian matrices, Vector Calculus – Gradient of Scalar Field, Directional Derivatives and normal vectors, Divergence of a vector field, curl of a vector, laplacian operator, Vector differentiation, Constrained optimization using Lagrange multipliers. Introduction to backpropagation, Backpropagation in neural networks . Self Study: SVM Lagrange multiplier Further Readings: Chain rule in Backpropagation		
UNIT 6	Optimization and Gradient Descent	08 HOURS
Applications/Case Study: Weather Forecasting Contents: STOCHASTIC PROCESSES AND MARKOV CHAINS, Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, nstep transition probabilities, Markov chain, Steady state condition, Markov analysis. Optimization basics in ML, Cost functions: MSE, cross-entropy, Derivatives and partial derivatives, Gradient Descent: Batch, Stochastic, Mini-batch, Learning rate and convergence, Convexity and local minima, Gradient in Backpropagation through Time(BPTT) Self Study: examples of Stochastic process and Markov chains Further Readings: RNN and LSTM Gradient Descent		

TEXT BOOKS

1. Jacqueline Kazil, Katharine Jarmul. (2016) *Data Wrangling with Python*. 1st Edition, O'Reilly Media, Inc. ISBN 9781491948774.
2. David S. Moore, George P. McCabe , Bruce A. Craig (2017). *Introduction to the Practice of Statistics*. 9th Edition, W.H.Freeman & Co Ltd. ISBN 978-1319013387.
3. Michael H. Kutner, Christopher J. Nachtsheim, John Neter (2019). *APPLIED LINEAR STATISTICAL MODELS*. 5th Edition, MCGRAWHILL EXCLUSIVE. ISBN 978-1259064746.
4. Douglas C. Montgomery, Cheryl L. Jennings , Murat Kulahci. *Introduction to Time Series Analysis and Forecasting*. 3rd Edition, Wiley. ISBN 978-1394186693.

REFERENCE BOOKS

1. Roy Jafari (2022). *Hands-On Data Preprocessing in Python: Learn how to effectively prepare data for successful data analytics*. 12th Edition, Packt Publishing. ISBN 978-1801072137.
2. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani (2021). *An Introduction to Statistical Learning: with Applications in R*. 1st Edition, 5th Printing (2017) Springer. ISBN 978-1461471370.
3. Tsay Ruey S. (2014). *Analysis of Financial Time Series*. 3rd Edition, Wiley India Pvt. Ltd. ISBN 9788126548934.
4. Anil Maheshwari (2023). *Data Analytics*. 2nd Edition, McGraw Hill India. ISBN 9789355324559.

E-RESOURCES

1. Data Analytics with Python, IIT Roorkee (https://onlinecourses.nptel.ac.in/noc21_cs45/preview)
2. Probability and Statistics, IIT Kharagpur (<https://nptel.ac.in/courses/111105090>)
3. MIT OpenCourseWare (<https://www.youtube.com/@mitocw/courses>)